Implement the Observer pattern in java

Program Description: Students must demonstrate the above pattern to observe the events happening in the stock market. Investors can subscribe to receive stock price updates for specific companies.

Method:

1. Subject: The subject maintains a list of observers and notifies them of state changes.
2. Observer: The observer interface defines the contract for concrete observer classes.
3. ConcreteSubject: A class that implements the subject interface and manages the observers.
4. ConcreteObserver: A class that implements the observer interface and receives notifications.

CODE:

```
import java.util.*;


// Main class (only public class)

public class ObserverPatternDemo {


    // Observer Interface

    interface Observer {

        void update(String stockName, double price);

    }


    // Subject Interface

    interface Subject {

        void subscribe(Observer o);

        void unsubscribe(Observer o);

        void notifyObservers();

    }


    // ConcreteSubject (Stock Market)
```

```
static class StockMarket implements Subject {
    private List<Observer> observers = new ArrayList<>();
    private String stockName;
    private double price;

    public void setStock(String stockName, double price) {
        this.stockName = stockName;
        this.price = price;
        notifyObservers();
    }

    public void subscribe(Observer o) {
        observers.add(o);
    }

    public void unsubscribe(Observer o) {
        observers.remove(o);
    }

    public void notifyObservers() {
        for (Observer o : observers) {
            o.update(stockName, price);
        }
    }
}

// ConcreteObserver (Investor)
static class Investor implements Observer {
    private String name;

    public Investor(String name) {
```

```
            this.name = name;
        }

        public void update(String stockName, double price) {
            System.out.println(name + " notified: " + stockName + " price is now " + price);
        }
    }

    // Main Method
    public static void main(String[] args) {
        StockMarket market = new StockMarket();

        Investor inv1 = new Investor("Alice");
        Investor inv2 = new Investor("Bob");

        market.subscribe(inv1);
        market.subscribe(inv2);

        market.setStock("TCS", 3500.50);
        market.setStock("INFY", 1450.75);

        market.unsubscribe(inv2);

        market.setStock("WIPRO", 420.30);
    }
}
```

OUTPUT:

Alice notified: TCS price is now 3500.5

Bob notified: TCS price is now 3500.5

Alice notified: INFY price is now 1450.75

Bob notified: INFY price is now 1450.75

Alice notified: WIPRO price is now 420.3